THE CYCLES ON AN OPEN SUBSCHEME, ITS COMPLEMENT AND THE AMBIENT SCHEME FIT INTO A RIGHT EXACT SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xypic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

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Allowed standard prerequisites (editorial).

- The group of k-cycles as the free abelian group on integral closed subschemes of δ-dimension k. Same chapter chow.tex, the definition of the group $Z_k(X)$ (used at proof lines 2278-2280). In the quasi-compact case it reduces the claim to a statement about the action of the two maps on a free basis.
- Pullback of cycles along an open immersion and pushforward along the complementary closed immersion. Same chapter chow.tex, the maps i_* and j^* of the sequence (statement lines 2266-2271, used at proof line 2281). They are the two arrows whose exactness is asserted, and the proof describes them on a basis element $[Z]$ by intersecting with U or vanishing when $Z \subset Y$.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 2264); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $U \subset X$ be an open subscheme, and denote $i : Y = X \setminus U \to X$ as a*

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reduced closed subscheme of X. For every $k \in \mathbf{Z}$ the sequence

$$Z_k(Y) \xrightarrow{i_*} Z_k(X) \xrightarrow{j^*} Z_k(U) \longrightarrow 0$$

is an exact complex of abelian groups.

Proof. First assume X is quasi-compact. Then $Z_k(X)$ is a free $\mathbf{Z}$-module with basis given by the elements $[Z]$ where $Z \subset X$ is integral closed of δ-dimension k. Such a basis element maps either to the basis element $[Z \cap U]$ or to zero if $Z \subset Y$. Hence the lemma is clear in this case. The general case is similar and the proof is omitted. $\square$